Gulp: A DevOps Based Tool for Web Applications

This article highlights the features of Gulp, an open source cross-platform streaming task runner that lets software developers automate many development tasks. It covers the installation of Gulp, and touches upon the code required for task and module management.

Software development teams need to work collaboratively on numerous tasks that are time consuming. DevOps is very popular in the corporate world for the automation, collaboration and real-time control that it enables on different types of software projects. Many corporate giants including Amazon, Oracle, Red Hat, SaltLake, Nagios, etc, use DevOps tools for real-world projects.

DevOps refers to the approach used to correlate the software development process with the various IT tasks to provide better software quality and streamlined deliveries. It speeds up the development, testing and deployment of code of high quality.

As the name indicates, DevOps is a combination of the development (Dev) and information technology operations (Ops) of an organisation that delivers software reliably, in a shorter time. The main purpose of DevOps is to automate the software development life cycle. The key features associated with DevOps are speed, reliability and better quality.

Agile development and enterprise systems management (ESM) make up the foundation of DevOps. Agile is linked to the software development process and ESM covers various processes that involve configuration management, system monitoring, etc. The need for DevOps arises because the operational team and development team work separately. The design, development and testing teams take more time and more errors occur during deployment.

The architecture of DevOps provides a combined approach for development and operations. This approach involves developing code, testing, planning, monitoring, deployment, operations and the release. This architecture is followed by large applications and those that are hosted on cloud platforms. This is because, in the case of large applications where the development and operational teams do not work in synchronised environments, long delays can occur in the process of design, deployment and testing. DevOps overcomes delays, maintaining high quality and timely delivery of the product.

The DevOps life cycle consists of seven phases which are: continuous development, continuous integration, continuous testing, continuous monitoring, continuous feedback, continuous deployment and continuous operations.

- **Continuous development:** This includes a plan for the software and development processes.
- **Continuous integration:** This consists of commit changes and integrated code from teams that have worked on the same project and have performed integration testing.
- **Continuous testing:** During this process, the team identifies bugs in the code with the help of automated testing, which is more reliable and less time consuming than manual testing.
- **Continuous monitoring:** This monitors all the processes that take place in different phases of the software development life cycle and records information regarding problems.
- **Continuous feedback:** This improves subsequent versions of the software by removing what is not relevant according to the customer.
- **Continuous deployment:** This involves code deployment on all the servers.
- **Continuous operation:** This is the automation of the process and key operations.